

Pre-Board Exam (2024-25)
Class: X
Subject: Artificial Intelligence (417)

Time: 2 hours

Max Marks: 50

General Instructions:

1. Please read the instructions carefully.
2. This Question Paper consists of 21 questions in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.
5. All questions of a particular section must be attempted in the correct order.
6. **SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**
 - i. This section has 05 questions.
 - ii. Marks allotted are mentioned against each question/part.
 - iii. There is no negative marking.
 - iv. Do as per the instructions given.
7. **SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**
 - i. This section has 16 questions.
 - ii. A candidate has to do 10 questions.
 - iii. Do as per the instructions given.
 - iv. Marks allotted are mentioned against each question/part.

SECTION A: OBJECTIVE TYPE QUESTIONS

Q. 1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4marks)	
i.	Shubham practises hard to get the best dancer award in the competition. What type of motivation is shown by Shubham? (a) Internal (b) External (c) Intermediate (d) None of the above	1
ii.	When you left click a file with a mouse it may get - (a) Selected b) Deselected c) Any of the above d) None of the above	1
iii.	Assertion(A): To communicate well in the class room is a natural ability. Reason(R): Effective teaching in the classroom demands knowledge of communication process. (a) Both A and R are correct and R is the correct explanation of A (b) Both A and R are correct but R is NOT the correct explanation of A (c) A is correct but R is not correct (d) A is not correct but R is correct	1
iv.	Society plays a role in boosting entrepreneurship by- (a) Providing raw material (b) Creating needs and demands (c) Both of the above (d) None of the above	1
v.	People coming from different areas have different perceptions, ideologies, and values. They often have less in common with someone from another part of the world. Those differences can affect their communication. This is an example of ____ (a) Interpersonal barrier b) Physical barrier (c) Cultural barrier d) Linguistic barrier	1
vi.	“Ensure equitable and inclusive quality education and promote lifelong learning opportunities for all”. Which SDG can you relate this statement to? (a) Life on land b) Clean water and sanitation (c) Quality education d) Reduced inequalities	1

vi.	F1 Score is the measure of the balance between (a) Accuracy and Precision (c) Recall and Accuracy b) Precision and Recall d) Recall and Reality	1
Q. 4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Sonali learnt about AI terminologies but was not able to recollect the term that is used to refer to machines that perform tasks with vast amounts of data using neural networks. Help her with the correct term. (a) NLP b) Computer Vision c) Deep learning d) Data Science	1
ii.	Statement1: Accuracy is the percentage of correct predictions out of all the predictions. Statement2: The real scenario is known as Prediction. (a) Both Statement1 and Statement2 are correct (b) Both Statement1 and Statement2 are incorrect (c) Statement1 is correct but Statement2 is incorrect (d) Statement2 is incorrect but Statement1 is correct	1
iii.	In which scenario would a high F1 Score be considered ideal? (a) High Precision and low Recall b) Low Precision and high Recall (c) High Precision and high Recall d) Low Precision and low Recall	1
iv.	Which of the following tasks identifies the object present in the image and where it is present in the image? (a) Classification b) Classification and localization (c) Object detection d) Instance segmentation	1
v.	_____ understands better the point of view in context to what is the expression behind the expressed opinion. (a) Virtual Assistants b) Sentiment Analysis (c) Text classification d) Automatic Summarization	1
vi.	Dinesh made an attendance system for the class based on face detection. System for which he had collected the data of all the students and used that data to train the model. Then, he used the same data to evaluate the model which resulted in the correct answer with all the students but not able to perform with unknown students. Name the concept.	1
Q. 5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	“Chatbot are used to solve the problems”. Identify the domain of AI used in it. (a) `Computer Vision b) Natural Language Processing (c) Data Science d) None of these	1
ii.	Which evaluation parameter can be defined as fraction of positive cases that are correctly identified.	1
iii.	What is semantic segmentation? (a) Categorizing each pixel in the image in to a specific class (b) Detecting the edges of the objects (c) Recognizing and tracking objects in videos (d) Labelling the image	1
iv.	One of the applications of NLP is relevant when used to provide overview of a news item or blog post, while avoiding redundancy from various sources and maximizing the diversity of content obtained. Identify the application from the following: (a) Sentiment Analysis b) Virtual assistant (c) Automatic summarization d) Text classification	1
v.	Which of the following is not a virtual assistant? (a) Alexa b) Cortana c) Silvi d) Siri	1
vi.	The result of comparison of Prediction and Reality can be recorded in what we call the_____ (a) Overfitting b) confusion matrix c) problem scoping d) data acquisition	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks) Answer each question in 20 – 30 words.

Q. 6	Write down the common communication barriers you may come across when you move to a new city or country.	2
Q. 7	What are the “SMART” goals which should be set in life to be successful?	2
Q. 8	Write the basic tips for taking care of laptops?	2
Q. 9	What are the myths related to Entrepreneur in society?	2
Q. 10	Mention any two human activities that can lead to the sustainable development.	2

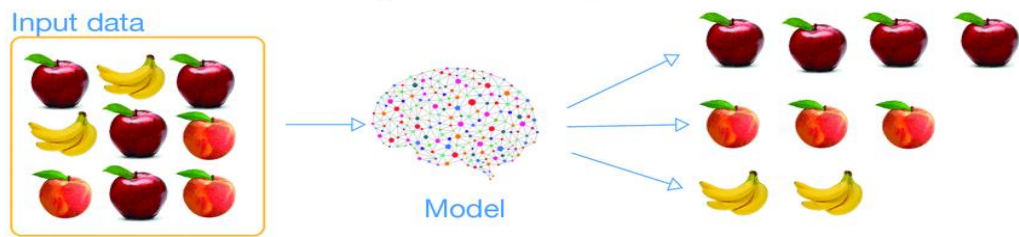
Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)

Q. 11	Justify the sentence – “For any AI project to be efficient, the training data should be authentic and relevant to the problem statement scoped.”	2
Q. 12	What are the advantages of Supervised learning over unsupervised learning in AI modeling.	2
Q. 13	Shailesh wants to access data from various sources for his AI model. Suggest him various ways of structured data collection from reliable sources.	2
Q. 14	Explain the term RGB image with an example.	2
Q. 15	Explain the two terms – stop words and bag of words.	2
Q. 16	Draw the confusion matrix for the following data • the number of true positive = 55 • the number of true negative = 20 • the number of false positive = 15 • the number of false negative = 10	2

Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)

Q. 17	Explain your friend about the various real life AI applications you use by giving a proper definition. Also help her understand the various ways how AI machines are trained.	4
Q. 18	People living at remote areas are facing the problem with medicine supplies. Can this problem be solved using any AI model? If it is feasible, prepare the following for it- (a) 4W Problem Canvas (b) Problem Statement Template	4
Q. 19	Identify and explain the types of the learning-based approaches in the figures given below. Figure 1	4

Figure 2



Q. 20

Normalize the given text and comment on the vocabulary before and after the normalization: “Raj and Vijay are best friends. They play together with other friends. Raj likes to play football but Vijay prefers to play online games. Raj wants to be a footballer. Vijay wants to become an online gamer.”

4

Q. 21

An AI model made the following prediction about the success of new product launched recently by a company:

Confusion Matrix		Reality	
		Yes	No
Prediction	Yes	55	25
	No	15	5

(i) Identify the total number of wrong predictions made by the model.

(ii) Calculate precision, recall and F1 Score.

4